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PUBLICATION LIST

PATENTS

- [P1] Reznik, L., & **Chuprov, S.** “Excluding Compromised Local Units from Receiving Global Parameters in Federated Learning Systems”, WIPO Publication No. [WO2024155819A1](#), published July 25, 2024.
Filed by Rochester Institute of Technology; priority date January 19, 2023.
- [P2] Reznik, L., & **Chuprov, S.** “Network Adjustment Based on Machine Learning End System Performance Monitoring Feedback”, WIPO Publication No. [WO2024059625A1](#), published March 21, 2024.
Filed by Rochester Institute of Technology; priority date September 14, 2022.

BOOKS & BOOK CHAPTERS

- [B1] **Chuprov, S.**, Zatsarenko, R., & Reznik, L. (2025) “mLINK: Machine Learning Integration with Network and Knowledge” in Metacognitive Artificial Intelligence, eds. Shakarian, P. & Wei, H. Cambridge University Press, September 2025, Chapter 16, pp. 264-275. doi: 10.1017/9781009522472. [URL](#)

REPRINTS

- [R1] **Chuprov, S.**, Belyaev, P., Gataullin, R., Reznik, L., Neverov, E., & Viksnin, I. (2024) “Robust Autonomous Vehicle Computer-Vision-Based Localization in Challenging Environmental Conditions” in Applied Sciences. 2024, pp. 34-48., doi: doi.org/10.3390/books978-3-7258-2184-6. *Reprint of the Special Issue “Challenges in the Guidance, Navigation and Control of Autonomous and Transport Vehicles” that was published in Applied Sciences in 2023.* [URL](#)

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- [J1] Chan, E. E. N., **Chuprov, S.**, Ovi, P. R., & Hasan, K. (2026). “A Granularity-Centered Taxonomy of Personalized Federated Learning” in Electronics. 2026, vol. 15, no. 15, 3481. doi: 10.3390/electronics15153481. [URL](#).
Published in Special Issue
- [J2] Korobeinikov, D., Zatsarenko, R., **Chuprov, S.**, Barea, A., & Reznik, L. (2026). “InteFL Framework: Optimizing Federated Learning with Metacognition for Application Design and Deployment” in IEEE Intelligent Systems. 2026. doi: 10.1109/MIS.2026.3658072. [URL](#). **Q1**. *Published in Special Issue*
- [J3] **Chuprov, S.**, Reznik, L., & Zatsarenko, R. (2025). “KISS: Knowledge Integration System Service for ML End Attacks Detection and Classification” in IEEE Transactions on Dependable and Secure Computing. 2025, pp. 1-12. doi: 10.1109/TDSC.2025.3553807. [URL](#). **Q1**.
- [J4] **Chuprov, S.**, Zatsarenko, R., Reznik, L., & Khokhlov, I. (2024). “Data Quality Based Intelligent Instrument Selection with Security Integration” in ACM Journal of Data and Information Quality. 2024, vol. 16, no. 3, pp. 1-24. doi: 10.1145/3695770. [URL](#). **Q2**.
- [J5] **Chuprov, S.**, Belyaev, P., Gataullin, R., Reznik, L., Neverov, E., & Viksnin, I. (2023). “Robust Autonomous Vehicle Computer-Vision-Based Localization in Challenging Environmental Conditions” in Applied Sciences. 2023, no. 13(9):5735. doi: 10.3390/app13095735. [URL](#). **Q1**. *Published in Special Issue*
- [J6] Berezovskaya, O., **Chuprov, S.**, Neverov, E., & Sadreev, E. (2023). “Review and Comparison of Lightweight Modifications of the AES Cipher for a Network of Low-Power Devices” in Automatic Control and Computer Sciences. 2022, no. 56, pp. 994-1006. doi: 10.3103/S0146411622080028. [URL](#)
- [J7] Melnikov, T., **Chuprov, S.**, Lazarev, E., Gataullin, R., & Viksnin, I. (2022). “Improving Reputation and Trust-Based Approach with Reliability Indicators for Autonomous Vehicles Intergroup Communication” in Tomsk State University Journal of Control and Computer Science. 2022, no. 61, pp. 108-117. [URL](#)
- [J8] Marinenkov, E., **Chuprov, S.**, Tursukov, N., Kim, I., & Viksnin, I. (2022). “Study on Destructive Informational Impact in Unmanned Aerial Vehicles Intergroup Communication” in Symmetry. 2022. Vol. 14, no. 8, pp. 1-18. [URL](#). **Q2**. *Published in Special Issue*

- [J9] Viksnin, I., Marinenkov, E., & **Chuprov, S.** (2022). “A Game Theory approach for communication security and safety assurance in cyber-physical systems with Reputation and Trust-based mechanisms” in Scientific and Technical Journal of Information Technologies, Mechanics and Optics, vol. 22, no. 1, pp. 47–59. doi: 10.17586/2226-1494-2022-22-1-47-59. [URL](#)
- [J10] Khokhlov, I., Reznik, L., & **Chuprov, S.** (2020). “Framework for Integral Data Quality and Security Evaluation in Smartphones” in IEEE Systems Journal, vol. 15, no. 2, pp. 2058-2065, doi: 10.1109/JSYST.2020.2985343. [URL](#). **Q1**
- [J11] **Chuprov, S.**, Viksnin, I., Kim, I., Tursukov, N., & Nedosekin, G. (2020). Empirical Study on Discrete Modeling of Urban Intersection Management System. International Journal of Embedded and Real-Time Communication Systems (IJERTCS), vol. 11(2), pp. 16-38, doi: 10.4018/IJERTCS.2020040102. [URL](#)
- [J12] **Chuprov, S.**, Viksnin, I., Kim, I., Marinenkov, E., Usova, M., Lazarev, E., Melnikov, T., & Zakoldaev, D. (2019). Reputation and Trust Approach for Security and Safety Assurance in Intersection Management System. Energies, 12(23), 4527, doi: 10.3390/en12234527. [URL](#). **Q1**

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- [C1] Amin, A., Hasan, K., Hong, L., **Chuprov, S.**, Thompson, R., Okpalaeze, A. D., Abena, P., Islam, T., & Ahmed, I. (2026). “Lightweight Privacy-First Federated Learning for Medical AI” in International Conference on Security and Privacy in Communication Systems (SecureComm 2026). doi: 10.1007/978-3-032-32764-2_13. [URL](#)
- [C2] **Chuprov, S.**, Lange, R. D., Reznik, L., Shakarian, P., Zatsarenko, R., & Korobeinikov, D. (2026). “Position: Artificial Intelligence Needs Meta Intelligence — the Case for Metacognitive AI” in Proceedings of the 43rd International Conference on Machine Learning (ICML 2026). [URL](#)
- [C3] Zatsarenko, R., Korobeinikov, D., **Chuprov, S.**, & Reznik, L. (2026). “Computationally Efficient Anomaly Detection and Exclusion for Practical and Robust Federated Learning” in IEEE/IFIP International Conference on Dependable Systems and Networks (DSN 2026).
- [C4] Peña, A. & **Chuprov, S.** (2026). “Improving PID-Based Aggregation in Federated Learning for Consumer Applications” in 2026 IEEE 23rd Consumer Communications & Networking Conference (CCNC). 2026. doi: 10.1109/CCNC65079.2026.11366291. [URL](#)
- [C5] Zatsarenko, R., **Chuprov, S.**, Korobeinikov, D., Reznik, L., & Peña, A. (2026). “How to Improve Federated Learning in Consumer Applications? Detect and Exclude Bad Clients” in 2026 IEEE 23rd Consumer Communications & Networking Conference (CCNC). 2026. doi: 10.1109/CCNC65079.2026.11366313. [URL](#)
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- [C9] Fonseca, P., **Chuprov, S.**, Solis, J. P., & Marquez, G. (2025). “Improving Lean Construction by Integrating ML for Contractor Performance Prediction” in 2025 38th Conference of Open Innovations Association (FRUCT). 2025. doi: 10.23919/FRUCT67853.2025.11239221. [URL](#)
- [C10] Chan, E. E. N. & **Chuprov, S.** (2025). “Bayesian Optimization of PID Coefficients for Robust Federated Learning” in TACCSTER 2025 Proceedings. 2025. doi: 10.26153/tsw/62178. [URL](#)
- [C11] Hong, L., **Chuprov, S.**, & Wan, Y. (2025) “Mitigating Health Misinformation Through Human-AI Collaboration” in AMCIS 2025 Proceedings. [URL](#)

- [C12] **Chuprov, S.**, Korobeinikov, D., Zatsarenko, R., & Reznik, L. (2025) “No trust, no learning? Improving Federated Learning Security and Robustness with Reputation and Trust” in Data Science Workshop “From Theory to Practice”. [URL](#)
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- [C14] Narayanan, A., **Chuprov, S.**, Reznik, L., Zatsarenko, R., & Korobeinikov, D. “Intelligent Soccer Event Detection and Highlights Generation with Broadcast Cues Integration” in 23rd International Conference on Machine Learning and Applications (ICMLA’24), Miami, FL, USA, 2024, pp. 554-559, doi: 10.1109/ICMLA61862.2024.00081. [URL](#)
- [C15] Patel, H., **Chuprov, S.**, Korobeinikov, D., Zatsarenko, R., & Reznik, L. (2024). “Improving Federated Learning Security with Trust Evaluation to Detect Adversarial Attacks” in 19th Annual Symposium on Information Assurance (ASIA’ 24), 2024, pp. 37-43. [URL](#)
- [C16] Korobeinikov, D., **Chuprov, S.**, Zatsarenko, R., & Reznik, L., (2024). “Federated Learning Robustness on Real World Data in Intelligent Transportation Systems” in 19th Annual Symposium on Information Assurance (ASIA’ 24), 2024, pp. 30-36. [URL](#)
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- [C23] Neverov, E., Viksnin, I., & **Chuprov, S.** (2023) “The Research of AutoML Methods in the Task of Wave Data Classification” in 2023 XXVI International Conference on Soft Computing and Measurements (SCM), 2023, pp. 156-158, doi: 10.1109/SCM58628.2023.10159058. [URL](#)
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- [C26] **Chuprov, S.**, Memon, M., & Reznik, L. (2023). “Federated Learning with Trust Evaluation for Industrial

- Applications” in 2023 IEEE Conference on Artificial Intelligence (CAI), Santa Clara, CA, USA, 2023, pp. 347-348, doi: 10.1109/CAI54212.2023.00153. [URL](#)
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- [C28] **Chuprov, S.**, Satam, A. N., & Reznik, L. (2022). “Are ML Image Classifiers Robust to Medical Image Quality Degradation?” in 2022 IEEE Western New York Image and Signal Processing Workshop (WNYISPW), 2022, pp. 1-4, doi: 10.1109/WNYISPW57858.2022.9983488. [URL](#)
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- [C33] **Chuprov, S.**, Gataullin, R., Neverov, E., Belyaev, P., Kim, I., & Viksnin, I. (2022). “Police Office Model Performance and Security Evaluation in a Simulated Group of Mobile Robots” in The 5th International Conference on Future Networks & Distributed Systems (ICFNDS 2021). Association for Computing Machinery, New York, NY, USA, pp. 606–615, doi: 10.1145/3508072.3508195. [URL](#)
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- [C37] Usova, M., Viksnin, I., & **Chuprov, S.** (2021). “Informational Messages and Space Models Application in Smart Factory Concept” in CEUR Workshop Proceedings, Vol. 2893, pp. 1-8. [URL](#)
- [C38] Khanh, T.D., Komarov, I., Don, L.D., Iureva, R., & **Chuprov, S.** (2020). “TRA: Effective Authentication Mechanism For Swarms Of Unmanned Aerial Vehicles” in 2020 IEEE Symposium Series on Computational Intelligence, pp. 1852-1858, doi: 10.1109/SSCI47803.2020.9308140. [URL](#)
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- [C44] **Chuprov, S., Viksnin, I., Kim, I. & Nedosekin, G.** (2019) “Optimization of Autonomous Vehicles Movement in Urban Intersection Management System” in 2019 24th Conference of Open Innovations Association (FRUCT), pp. 60-66, doi: 10.23919/FRUCT.2019.8711967. [URL](#)
- [C45] **Chuprov, S.**, Viksnin, I., Kim, I., & Usova, M. (2019). “Intersection Management Tasks in Mobile Robotic System with Decentralized Control” in CEUR Workshop Proceedings, vol. 2344, pp. 1-12. [URL](#)
- [C46] Usova, M., **Chuprov, S.**, Viksnin, I., Gataullin, R., Komarova, A., & Iuganson, A. (2019). “Model of Smart Manufacturing System” in International Symposium on Intelligent and Distributed Computing, pp. 356-362, doi: 10.1007/978-3-030-32258-8_42. [URL](#)
- [C47] Marinenkov, E., **Chuprov, S.**, Viksnin, I., & Kim, I. (2019). “Empirical Study on Trust, Reputation, and Game Theory Approach to Secure Communication in a Group of Unmanned Vehicles” in CEUR Workshop Proceedings, vol. 2590, pp. 1-12. [URL](#)
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- [C49] Viksnin, I., Lyakhovenko, J., Tursukov, N., **Chuprov, S.**, & Sozinova, E. (2019). “Empirical Study on Modeling of People Behavior in Emergency” in CEUR Workshop Proceedings, vol. 2590, pp. 1-8. [URL](#)
- [C50] Kim, I., Matos-Carvalho, J. P., Viksnin, I., Campos, L. M., Fonseca, J. M., Mora, A., & **Chuprov, S.** (2019). “Use of Particle Swarm Optimization in Terrain Classification based on UAV Downwash” in 2019 IEEE Congress on Evolutionary Computation (CEC), pp. 604-610, doi: 10.1109/CEC.2019.8790031. [URL](#)
- [C51] Viksnin, I., **Chuprov, S.**, Usova, M., & Zakoldaev, D. (2019). “Police Office Model for Multi-agent Robotic Systems” in IOP Conference Series: Materials Science and Engineering, vol. 497, no. 1, p. 012036, doi: 10.1088/1757-899X/497/1/012036. [URL](#)

DISSERTATIONS

- [D1] **Chuprov, S.** (2024) “Robust Machine Learning Under Vulnerable Cyberinfrastructure and Varying Data Quality”, Ph.D. dissertation, B. Thomas Golisano College of Computing and Information Sciences, Rochester Institute of Technology, Rochester, NY, USA, 2024. [URL](#)
- [D2] **Chuprov, S.** (2019) “Assurance of Information in a Mobile Robotic System with Decentralized Control”, Master’s dissertation, Department of Secure Information Technologies, ITMO University, Saint Petersburg, Russia, 2019.

OTHER PUBLICATIONS AND PRESENTATIONS

- [O1] Fonseca, P., & **Chuprov, S.** (2025). “SPAM Email Detection App Using Machine Learning Techniques” in *UTRGV 2025 STEM Research Conference*. Presented May 2, 2025. — *Best Doctoral Poster Presentation in CS*
- [O2] Belov, A., Belyaev, P., Viksnin, I., Kim, I., Radabolsky, V., Turushev, T., & **Chuprov, S.** (2023). “Software and Hardware Bundle for Controlling a Group of Unmanned Vehicles Based on Robust Computer Vision Algorithms” in LETI Transactions on Electrical Engineering & Computer Science. 2023, vol. 16, no. 6. pp. 52–69. — *in Russian*. [URL](#)
- [O3] Berezovskaya, O., **Chuprov, S.**, Neverov, E., & Sadreev., E. (2022). “Review and Comparison of AES Lightweight Modifications for a Low-Power Devices Network” in Problems of Information Security. Computer Systems. no. 2, pp. 35-50. doi: 10.48612/jisp/gp9v-96dh-32
v1. — *in Russian*. [URL](#)
- [O4] Buzina, E., **Chuprov, S.**, Tursukov, N., Belyaev, P., & Viksnin, I. (2022). “Algorithm for Uniform Coverage of the Monitoring Territory by Unmanned Aerial Vehicles” in LETI Transactions on Electrical Engineering &

- [O5] **Chuprov, S.**, Gataullin, R., & Viksnin, I. (2021). “Vulnerabilities Assessment in Mobile Robotic Control System” in Proceedings of the St. Petersburg State Electrotechnical University LETI, vol. 10, pp. 70-75. – *in Russian*.
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- [O6] **Chuprov, S.** (2020). “Intersection Management Tasks Application in Mobile Robotic System with Autonomous Vehicle Models” in Saint-Petersburg’s Government Grant Award Winners, pp. 68-74. [URL](#)
- [O7] **Chuprov, S.** (2020). “Reputation, Trust, and Data Quality Concept for Security and Safety Assurance in a Group of Autonomous Vehicles” in IX Young Scientists Congress Proceedings, Electronic Edition. [URL](#)
- [O8] **Chuprov, S.** (2019). “Security Assurance of Mobile Robotic Systems with Decentralized Control” in 2019 ITMO University Annotated Master’s Research Thesis Proceedings, pp. 66-72. [URL](#)
- [O9] **Chuprov, S.**, Viksnin, I., & Kim, I. (2019). “Organization of Safe Intersections Passage by Unmanned Vehicles in a Mobile Robotic System” in VIII Young Scientists Congress Proceedings, Electronic Edition. [URL](#)
- [O10] **Chuprov, S.**, Viksnin, I., Kim, I., & Usova, M. (2019). “Secure Intersection Management Passage with a Group of Autonomous Vehicles in Mobile Robotic System with Decentralized Control” in ITMO University Young Scientists Almanac, vol. 2, pp. 43-48. [URL](#)
- [O11] **Chuprov, S.**, Viksnin, I., & Usova, M. (2018). “Police Office Model with Partial Decentralized Control in Mobile Robotic System” in Regional Informatics and Information Security SPOISU Proceedings, vol. 6, pp. 245-249.
[URL](#)

Presentations, posters, forums, workshops, and seminars w/o publications:

- [NP1] Fonseca, P., & **Chuprov, S.** (2025). “SPAM Email Detection App Using Machine Learning Techniques” in *STARTER-AI Summer 2025 Workshop*. Presented May 21, 2025.
- [NP2] **Chuprov, S.** (2024). “About my Research, Approach to Mentorship, and Some Advice from my Fresh Grad Student’s Perspective” at First-Year PhD Students in Computer Science seminar at UTRGV, 16 September 2024, Edinburg, TX, USA.
- [NP3] **Chuprov, S.** (2024). “Security and Robustness of Machine Learning under Vulnerable Cyberinfrastructure and Varying Data Quality” at [From Theory to Practice \(T2P\)](#) Workshop, 15-19 April 2024, Kigali, Rwanda – *Remote presentation*.
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